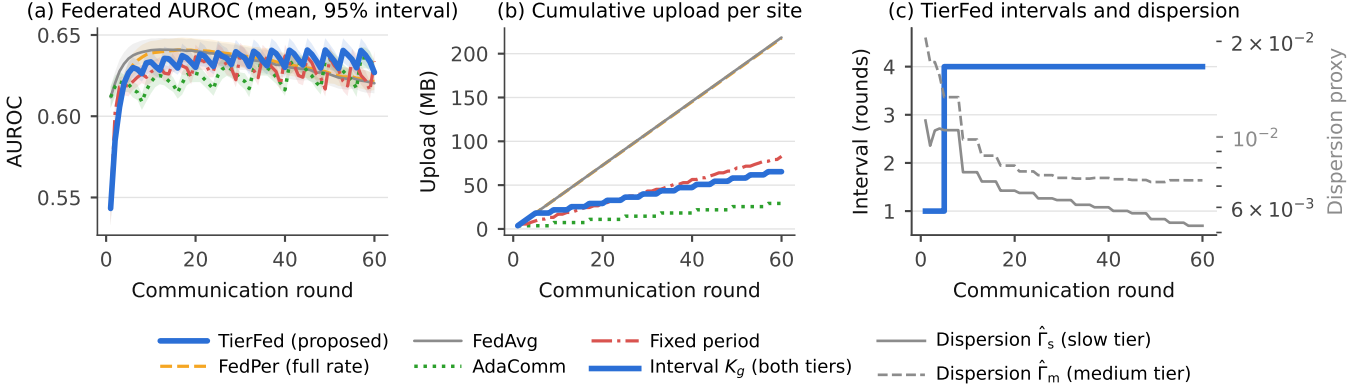
(a) Federated AUROC (mean, 95% interval)
AUROC
0.65
0.60
0.55
0
20
40
60
Communication round
(b) Cumulative upload per site
Upload (MB)
200
150
100
50
0
0
20
40
60
Communication round
(c) TierFed intervals and dispersion
Interval (rounds)
4
3
2
1
2×10^{-2}
10^{-2}
6×10^{-3}
Dispersion proxy
0
20
40
60
Communication round
TierFed (proposed)
FedPer (full rate)
FedAvg
AdaComm
Fixed period
Interval K_g (both tiers)
Dispersion $\hat{\Gamma}_s$ (slow tier)
Dispersion $\hat{\Gamma}_m$ (medium tier)